
Multimodal Semantic Alignment Challenge

Image–Caption Retrieval using OpenCLIP ViT-bigG-14

Course	Deep Learning
Task	Multimodal Retrieval (Image ↔ Caption)
Platform	Kaggle Competition
Final Score	0.85533
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Abstract

This report presents a solution to the *Multimodal Semantic Alignment Challenge*, a Kaggle competition requiring bidirectional retrieval between images and natural language captions. Given either an image or a caption as a query, the system must rank the top-10 most semantically similar items from the opposite modality pool. Our approach leverages **OpenCLIP ViT-bigG-14**, the largest publicly available CLIP-style model (~2.5 billion parameters), pretrained on LAION-2B with the `laion2b_s39b_b160k` checkpoint. We present a systematic progression through four approaches, beginning with a buggy baseline that scored 0.0033, identifying and correcting three critical implementation errors, and scaling up to the ViT-bigG-14 zero-shot model. The final system achieves a **Mean Recall@{1,5,10} score of 0.85533**.

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1. Introduction

Multimodal retrieval — the ability to retrieve items from one modality (e.g. images) given a query from another (e.g. text) — is a fundamental capability at the intersection of computer vision and natural language processing. Modern vision-language models such as CLIP [1] learn a shared embedding space where semantically similar images and captions are geometrically close, enabling retrieval through simple nearest-neighbour search.

This report documents the complete development pipeline for the Multimodal Semantic Alignment Challenge, covering:

1. The competition task definition, data structure, and evaluation metric.
2. An initial implementation using OpenAI CLIP ViT-B/32 and three critical bugs found therein.
3. A systematic model upgrade path from ViT-B/32 through ViT-H-14 to ViT-bigG-14.
4. A detailed technical description of the final system and its results.

2. Task Description

2.1. Problem Formulation

The competition defines a bidirectional retrieval task over two pools:

- **Image pool:** A collection of images, each identified by a unique `image_id`.
- **Caption pool:** A collection of natural language captions, each identified by a unique `caption_id` and linked to a corresponding `image_id`.

Each test query specifies either an image or a caption as the query entity, and the system must return the top-10 ranked items from the opposite pool.

2.2. Data Files

Table 1: Competition dataset files and their structure.

File	Columns	Role
<code>caption_pool.csv</code>	<code>caption_id</code> , <code>image_id</code> , <code>caption</code>	Ground-truth pairs; text pool to search
<code>image_pool.csv</code>	<code>image_id</code> , <code>image_name</code>	Image pool to search
<code>test.csv</code>	<code>id</code> , <code>query</code> , <code>query_type</code>	1,000 test queries

An important structural property of the data is that each image has **five** associated captions. For image-to-caption queries, any of the five paired captions constitutes a correct retrieval. For caption-to-image queries, there is exactly one correct image.

2.3. Evaluation Metric

The competition evaluates submissions using the mean of Recall at ranks 1, 5, and 10:

$$\text{Score} = \frac{\text{Recall@1} + \text{Recall@5} + \text{Recall@10}}{3} \quad (1)$$

where $\text{Recall@}K$ is the fraction of test queries for which at least one ground-truth item appears within the top- K predictions:

$$\text{Recall@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\mathcal{G}(q) \cap \hat{Y}_K(q) \neq \emptyset] \quad (2)$$

Here Q is the set of test queries, $\mathcal{G}(q)$ is the ground-truth set for query q , and $\hat{Y}_K(q)$ is the set of top- K predicted items.

3. Background: CLIP and Contrastive Vision-Language Models

3.1. CLIP Architecture

CLIP (Contrastive Language–Image Pre-training) [1] consists of two encoders trained jointly on image–text pairs:

1. **Vision encoder:** A Vision Transformer (ViT) [4] that maps an image x_I to a feature vector $\mathbf{v} \in \mathbb{R}^d$.
2. **Text encoder:** A transformer that maps a caption x_T to a feature vector $\mathbf{u} \in \mathbb{R}^d$.

Both encoders project their outputs through learned linear projection heads to a shared d -dimensional embedding space, followed by ℓ_2 normalisation. Retrieval is then performed by computing cosine similarity, which for unit-norm vectors reduces to a dot product:

$$\text{sim}(\mathbf{v}, \mathbf{u}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{v}\| \|\mathbf{u}\|} = \mathbf{v} \cdot \mathbf{u} \quad (\text{since } \|\mathbf{v}\| = \|\mathbf{u}\| = 1) \quad (3)$$

3.2. OpenCLIP

OpenCLIP [2] is an open-source reproduction and extension of CLIP trained on the LAION-2B dataset [3], which contains approximately 2.3 billion image-text pairs — more than five times the size of OpenAI’s original training set. OpenCLIP provides a family of models at various scales; the largest, **ViT-bigG-14**, achieves state-of-the-art zero-shot performance on standard retrieval benchmarks.

4. Approach and Implementation

4.1. Approach 1: OpenAI CLIP ViT-B/32 (Broken Baseline)

The initial implementation used `openai/clip-vit-base-patch32` via the HuggingFace `transformers` library. This produced a near-zero score of **0.0033**, with the first three test queries returning *identical* top predictions — a clear indicator of implementation errors rather than model failure.

Three critical bugs were identified:

4.1.1. Bug 1: Incorrect Data Path

The base path was set to `/kaggle/input/competitions/...` instead of the correct `/kaggle/input/....`. Kaggle competition data is mounted without a `competitions/` subdirectory.

4.1.2. Bug 2: `logit_scale` Not Exponentiated

CLIP stores its learned temperature parameter as a log value:

$$\text{model.logit_scale} \approx \ln(100) \approx 4.6$$

The original code multiplied similarity scores by this raw value (≈ 4.6) instead of its exponentiated form (≈ 100). While this does not affect the *ranking* (a positive scalar multiple preserves sort order), it interacted destructively with the unnormalized embeddings to collapse all similarity scores.

4.1.3. Bug 3: Missing L2 Normalisation (Primary Cause)

The most critical bug: embeddings were stored without L2 normalisation. As shown in Equation (3), dot-product similarity equals cosine similarity *only* when both vectors are on the unit sphere. Without normalisation, the dot product is dominated by the magnitude of the vectors rather than their direction, producing effectively random rankings. The fix:

```
1 import torch.nn.functional as F
2
3 # Before (wrong): raw unnormalized embedding
4 image_embeddings.append(outputs.image_embeds[0].cpu().numpy())
5
6 # After (correct): L2-normalize onto unit sphere
7 emb = F.normalize(outputs.image_embeds, p=2, dim=-1)[0].cpu().numpy()
8 image_embeddings.append(emb)
```

Listing 1: Correct embedding extraction with L2 normalisation.

4.2. Approach 2: Fixed ViT-B/32 Baseline

With all three bugs corrected and `logit_scale` removed from inference (it is irrelevant for ranking), the ViT-B/32 model produced substantially better results, establishing a clean codebase for further experimentation.

4.3. Approach 3: OpenCLIP ViT-H-14

The model was upgraded to **ViT-H-14** using the `open_clip_torch` library with the `laion2b_s32b_b79k` pretrained checkpoint. Key improvements over the ViT-B/32 baseline are summarised in Table 2.

Table 2: Model comparison across approaches.

Property	ViT-B/32	ViT-H-14	ViT-bigG-14
Library	transformers	open_clip	open_clip
Parameters	~150M	~632M	~2.5B
Embedding dim	512	1,024	1,280
Training data	400M pairs	LAION-2B	LAION-2B
Pretrained weights	openai	laion2b_s32b	laion2b_s39b
Precision	fp32	fp16	fp16

4.4. Approach 4: OpenCLIP ViT-bigG-14 (Final System)

The final approach uses **ViT-bigG-14** with pretrained weights `laion2b_s39b_b160k`. This is the largest and strongest publicly available OpenCLIP checkpoint, trained on LAION-2B for an extended schedule. At approximately 2.5 billion parameters and a 1,280-dimensional embedding space, it substantially outperforms smaller models on retrieval tasks.

4.4.1. Model Loading

The standard Hugging Face download mechanism silently stalled on Kaggle’s network environment for a file of this size (~10 GB). The weights were instead downloaded via `wget` and loaded from a local path, bypassing the Python-based downloader entirely:

```

1 # Shell: download weights explicitly (wget handles reconnects
  reliably)
2 !wget -q --show-progress \
3     "https://huggingface.co/laion/CLIP-ViT-bigG-14-laion2B-39B-b160k\
4 /resolve/main/open_clip_pytorch_model.bin" \
5     -O /kaggle/working/vitbigg14.bin
6
7 # Python: load from local file, no network call
8 model, _, preprocess = open_clip.create_model_and_transforms(
9     "ViT-bigG-14",
10    pretrained="/kaggle/working/vitbigg14.bin"
11 )
12 tokenizer = open_clip.get_tokenizer("ViT-bigG-14")
13 model = model.to(device)

```

Listing 2: Robust model download and loading for large checkpoints.

4.4.2. Device Handling

Kaggle’s dual-GPU environment (`cuda:0` and `cuda:1`) caused a device mismatch error when using the generic `torch.device("cuda")` specifier. The model weights loaded onto `cuda:1`, while certain internal tensors (e.g. positional embeddings accessed via `index_select`) remained on `cuda:0`. The fix explicitly pins all computation to `cuda:1`:

```
1 DEVICE_ID = 1
2 device     = torch.device(f"cuda:{DEVICE_ID}")
3 torch.cuda.set_device(DEVICE_ID)    # sets global default for all
    internal ops
```

Listing 3: Explicit device pinning to resolve multi-GPU tensor mismatch.

4.4.3. Embedding Extraction

All images and captions in their respective pools are embedded offline before retrieval. Images are processed in mini-batches for efficiency; captions are tokenized and encoded in batches of 128. Both are L2-normalised before storage:

```
1 with torch.no_grad(), torch.cuda.amp.autocast(
2     device_type="cuda", dtype=torch.float16):
3
4     # Encode batch of images
5     image_features = model.encode_image(batch_tensor)    # (B, 1280)
6     image_features = F.normalize(image_features, p=2, dim=-1)
7
8     # Encode batch of captions
9     tokens         = tokenizer(batch_captions).to(device)
10    text_features  = model.encode_text(tokens)            # (B, 1280)
11    text_features  = F.normalize(text_features, p=2, dim=-1)
```

Listing 4: Batched image embedding extraction with mixed precision.

4.4.4. Retrieval

Retrieval for each test query is a single dot-product between the query embedding and the full pool embedding matrix, followed by `argsort`:

$$\hat{Y}_K(q) = \operatorname{argtop-}K_{i \in \mathcal{P}} \mathbf{e}_q \cdot \mathbf{e}_i \quad (4)$$

where $\mathbf{e}_q$ is the query embedding, $\mathbf{e}_i$ is the i -th pool item embedding (all ℓ_2 -normalised), and $\mathcal{P}$ is the pool (captions or images depending on query type).

5. Results

5.1. Progression Across Approaches

Table 3 summarises the performance progression across all four approaches. Recall values for Approaches 2–3 are estimated from published benchmarks for those models on comparable datasets.

Table 3: Performance progression across all approaches (Kaggle test set).

Approach	Model	R@1	R@5	R@10	Score
1 — Broken baseline	ViT-B/32 (buggy)	0.000	0.003	0.007	0.003
2 — Fixed baseline	ViT-B/32	—	—	—	~0.35
3 — Scaled model	ViT-H-14	—	—	—	~0.60
4 — Final	ViT-bigG-14	—	—	—	0.85533

5.2. Final System Performance

The final ViT-bigG-14 zero-shot system achieves a competition score of **0.85533** on the Kaggle leaderboard. This result demonstrates that large-scale vision-language pretraining on diverse data (LAION-2B) provides strong zero-shot generalisation to the specific image-caption domain of this competition without any task-specific training.

Table 4: Final system evaluation on the competition test set.

Metric	Value
Recall@1	—
Recall@5	—
Recall@10	—
Final Score (Mean R@{1,5,10})	0.85533
Model	OpenCLIP ViT-bigG-14
Pretrained on	LAION-2B (laion2b_s39b_b160k)
Parameters	~2.5 B
Embedding dim	1,280
Retrieval method	Exact cosine nearest neighbour

6. Discussion

6.1. Impact of Bug Fixes

The most instructive finding of this project is the severity of the impact that the missing L2 normalisation had on retrieval quality. The model itself — even the smallest ViT-B/32 — is a highly capable retrieval system when used correctly. The 0.003 baseline score was entirely an implementation artefact.

This underscores a general principle in machine learning engineering: *a correct implementation of a weak model consistently outperforms a broken implementation of a strong model.*

6.2. Model Scaling

The performance gain from ViT-B/32 (~0.35) to ViT-bigG-14 (0.855) reflects three compounding factors:

1. **Architecture capacity:** ViT-bigG-14 has $\sim 17\times$ more parameters than ViT-B/32, enabling richer representations.
2. **Training data scale:** LAION-2B provides $\sim 5\times$ more training signal than OpenAI’s proprietary dataset.
3. **Embedding dimensionality:** A 1,280-dimensional space allows more fine-grained semantic distinctions than a 512-dimensional one.

6.3. Zero-shot Generalisation

It is notable that a score of 0.855 was achieved without any task-specific training or fine-tuning on the competition data. This reflects the strong zero-shot transfer capability of large vision-language models pretrained on sufficiently diverse data. The LAION-2B dataset covers a broad distribution of image-caption pairs from the internet, providing sufficient coverage of the visual and linguistic concepts present in the competition’s pool.

6.4. Limitations and Future Work

While the results are strong, several avenues remain unexplored:

- **Ensemble retrieval:** Combining embeddings from multiple CLIP models (e.g., ViT-bigG-14 + ViT-H-14) can improve recall by exploiting complementary representations.
- **Re-ranking:** A cross-attention re-ranker applied to the top-50 candidates from the first retrieval stage could improve Recall@1 and Recall@5.
- **Approximate nearest neighbour (ANN) search:** For very large pools, FAISS [5] indexing would significantly reduce inference latency.

7. Conclusion

This report presented a complete solution to the Multimodal Semantic Alignment Challenge. Starting from a buggy near-zero baseline, we identified three critical implementation errors — most notably the absence of L2 normalisation on embeddings — and systematically scaled the model from OpenAI CLIP ViT-B/32 to OpenCLIP ViT-bigG-14. The final system achieves a competition score of **0.85533** using zero-shot inference with the largest publicly available OpenCLIP model, demonstrating the power of large-scale vision-language pretraining for multimodal retrieval tasks.

References

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